

Generation Interconnection System Impact Study Report

For Queue Project Q23-2366

Skiland 138 kV

April 17, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2366. The Interconnection Customer proposes a 128.5 MW battery energy storage facility interconnecting at the Skiland 138 kV substation in the Pennsylvania area. The request was received on December 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 466 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,321,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Skiland 500 kV/345 kV TX
Contingency: Deland W 500/345kV XFMR
Loading: 110.2% (Rating: 781 MVA)
DFAX: 43.6%
Pre-project loading: 103.0%
- Facility: Gilstrap - Stoner 230 KV line
Contingency: El Paso - Ravenna 500 KV
Loading: 106.5% (Rating: 578 MVA)
DFAX: 5.5%
Pre-project loading: 105.2%
- Facility: Ripon Ind Park - Joseph J Seymour Power Project 138 KV
Contingency: Corning - Skiland 345 KV
Loading: 104.1% (Rating: 206 MVA)
DFAX: 10.8%
Pre-project loading: 97.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Skiland 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$28,936,000	28% (\$7,977,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gilstrap - Stoner 230 KV line	Reconductor line	\$28,305,000	4% (\$1,060,000)
Ripon Ind Park - Joseph J Seymour Power Project Substation 138 KV	Reconductor line	\$1,084,000	26% (\$284,000)
	Total allocated to project		\$9,321,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.